

CUSP Longitudinal Health AI Proposal

Yihan and the MD6204 project team
14 September 2026

CUSP connects wearable signals, lifestyle context and glucose measurements in one personal timeline. Its translation strategy links consumer metabolic wellness to authorised professional follow-up through a shared longitudinal model. An initial technical PoC and working product prototype now provide a concrete basis for testing measurement burden, personalisation and reliable interpretation.

Explore CUSP: [Open the interactive Figma app](#) | [View the Canva project materials](#)

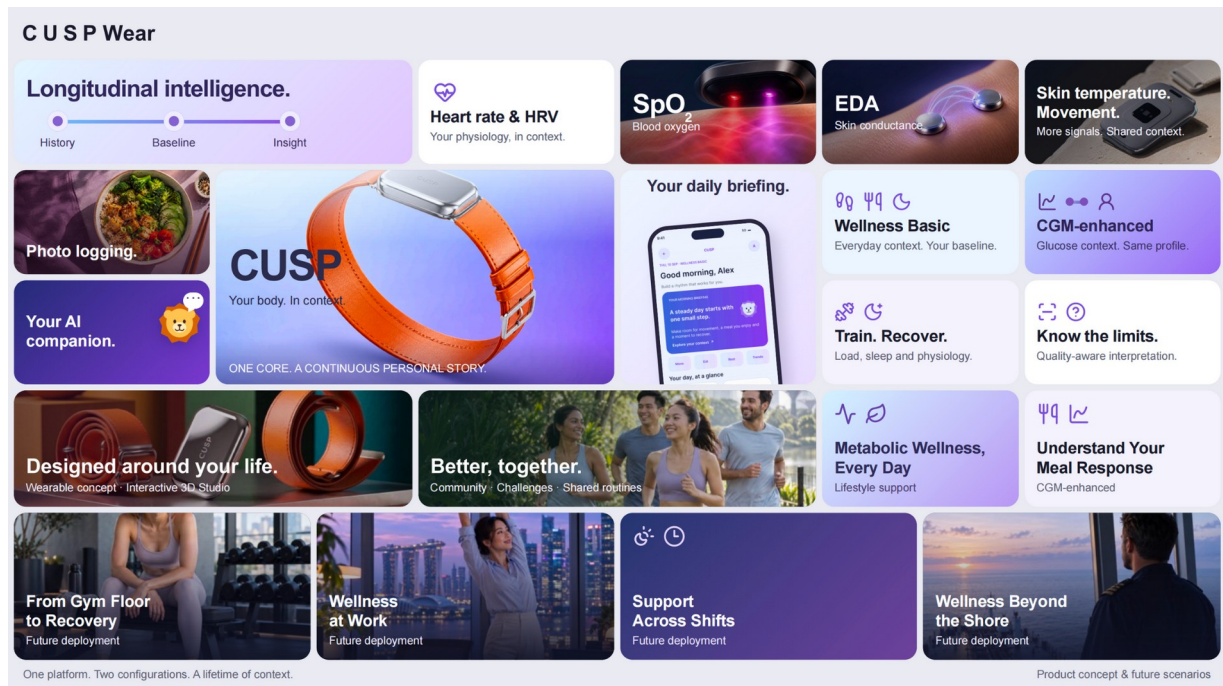


Figure 1. CUSP product vision. Hardware, sensing capabilities including SpO2, and future deployment settings are concepts, not validated capabilities of the current PoC. Initial research can use compatible third-party devices.

From consumer engagement to professional follow up

ToC consumer value. Build a sustained user relationship through daily briefings, logging and lifestyle support. Wellness Basic provides baseline context without numerical glucose estimates; CGM-enhanced adds measured glucose and meal-response context. The value to test is lower measurement burden while retaining useful metabolic insight.

Shared intelligence core. A personal model and proposed longitudinal care record (ECR) organise consented measurements, events and baseline changes into a structured trajectory for personal insight and authorised professional review.

ToB professional value. Start with professional longitudinal summaries; then validate follow-up prioritisation and post-discharge monitoring. Readmission research and future escalation or care-support workflows extend the platform roadmap and require separate clinical evidence.

An existing product foundation for translation

CUSP already has an initial technical PoC, a clickable English app, a working local frontend, defined user workflows, wearable 3D concepts, documented safety boundaries, an initial regulatory strategy and a preliminary commercial model. The next investment can test specific scientific and deployment uncertainties using this existing product foundation. [2]

Product boundaries separate wellness context from glucose estimates, address insufficient data and retain professional decisions. Clinical validity, sensor performance and formal regulatory classification remain to be established.

Research Evidence and Validation Plan

Sparse measurements to longitudinal personal intelligence

Initial technical evidence

The team's TEFN study evaluates 30-minute glucose forecasting from approximately two hours of history, with shared HR, activity, EDA, temperature, meal and time-of-day features. It reports 3,799 test windows across 15 participants, five seeds per condition and 2,000 participant-cluster bootstrap iterations. [1]

Glucose input in the history window	RMSE mg/dL	Increase vs dense input
24 points with a current measurement	14.371	Reference
4 points at t, t-30, t-60 and t-90 min	15.322	6.63%
4 points with the latest 15 minutes old	17.810	23.95%
No glucose input	21.631	50.54%

Recency matters more than density in this experiment. The four-point current-input condition has a paired 95% confidence interval of 2.84–10.18% for the RMSE increase; no sparse condition met the prespecified 10% noninferiority criterion. Older measurements cannot simply be assumed to substitute for current glucose.

The study does not establish CGM replacement or useful glucose forecasts weeks after sensor removal. The public BIG IDEAs source cohort spans about 8–10 days; testing history across distant episodes requires prospective longitudinal data. These are team-reported retrospective results, with reproduction and causal information-flow checks planned. [1, 5]

How little high quality measurement is enough

The next question is whether personal history adds value beyond recent inputs: can earlier paired episodes improve task-specific interpretation as current measurements become sparse? Full CGM should remain hidden evaluation ground truth when it is not an allowed input. History benefit must be demonstrated on held-out evaluation.

Uncertainty and abstention connect model performance to product behaviour. Test calibrated prediction intervals, missingness and temporal shift; measure error as unreliable outputs are withheld. Confidence intervals around aggregate RMSE do not establish calibrated confidence for an individual prediction.

Proposed 18 month programme

Months 1 to 3. Reproduce the density and recency results; audit causal preprocessing and participant/temporal splits. Establish transparent baselines, input-availability rules and prespecified task-specific success criteria with the PI.

Months 4 to 9. Vary anchor timing, history length and sensor dropout while holding current information constant. Deliver measurement-burden versus performance curves, history ablations, calibration and abstention analyses.

Months 10 to 18. Subject to approvals and access, collect longer paired episodes and run consumer pilots. Evaluate cross-episode benefit, adherence, burden and explanation understanding. Test professional summary completeness, provenance and review time with a suitable partner.

Connecting the research to value

Consumer value to validate. Lower measurement burden while retaining useful metabolic insight.

Professional value to validate. Structured trajectories for professional review, followed by separate validation of prioritisation.

Programme outcome. A reproducible benchmark, validated output limits, prospective pilot evidence and a use-case investment decision.

Singapore Deployment and Commercialisation

An existing product with specific risks to resolve

Why Singapore and why NTU

CUSP's deployment thesis is Singapore as a high-density translational testbed. LKCMedicine and NHG jointly established a primary-care research centre; Healthier SG connects family-doctor relationships, preventive care and community support. This ecosystem offers a basis for testing consumer engagement and professional review locally. [3]

The proposed route begins with NTU-based scientific leadership and potential LKCMedicine clinical collaboration, then campus, gym, employer and occupational-health pilots. Community, primary-care and referral partners could support prospective paired-data collection and authorised professional follow-up studies. Access, ethics approvals and partner commitments must be secured; no institutional endorsement or deployment agreement is assumed.

Consumer longitudinal engagement → prospective paired real-world data → professional trajectory summaries → follow-up and post-discharge validation. Singapore is the proposed setting in which to test this sequence, with success assessed at each stage before expanding.

From ToC engagement to ToB institutional value

Consumer subscriptions and an optional CGM-enhanced service are the initial revenue hypotheses. Daily routines and Community support sustained use and repeat interaction; consented longitudinal history can improve personal baselines and, if validated, create value beyond a single app session.

The same longitudinal core and proposed ECR record can support professional summaries, then studies of follow-up prioritisation and post-discharge monitoring. Readmission research and future escalation or care-support workflows define the longer-term platform opportunity. These require separate clinical evidence and regulatory assessment; AI would not independently diagnose, change treatment or decide admission.

Potential ToB models include programme fees, per-patient services and institutional licensing. Early professional validation should establish review utility and a credible buyer workflow before testing these models. Defensibility depends on demonstrated history benefit, traceability and workflow adoption.

What the funding is intended to de risk

Remaining uncertainty	Funded work and decision evidence
Scientific value	Longer paired data and controlled history studies: does history preserve useful insight as references age, and when must the system abstain?
Deployment and workflow	Device and frontend integration, consumer pilots and professional review studies: adherence, burden, comprehension, summary completeness and review time.
Commercial viability	Test subscription retention, willingness to pay and partner use cases; develop a costed delivery model and an evidence-based market entry decision.
Clinical and institutional readiness	Scope intended uses, consent and data governance; assess IP and regulatory pathways; agree a follow-on clinical protocol with qualified partners.

Funding advances a defined product and initial PoC by testing reliable inference, sustained use, professional utility and deployment viability.

The NTUitive Gap Fund offers up to S\$250,000 over 18 months, with the current call closing on 18 September 2026. We seek academic PI leadership, clinical advisors and pilot partners to scope a credible application. The costed request, eligibility, team roles and IP arrangements must be agreed; S\$250,000 is the fund ceiling, not an approved project budget. [4]

Evidence and sources: [1] MD6204 Project Official Doc, pp. 94–98; TP2 merged working document. [2] CUSP Master Story v2.2 ToC ToB, product assets and safety review. [3–5] Official ecosystem, funding and dataset sources below.

[\[3\] NTU and NHG research centre](#) | [\[3\] Healthier SG](#) | [\[4\] NTUitive call](#) | [\[5\] BIG IDEAs dataset](#)